

COSC 31093 /BECS 31213 - Enterprise Software Design and Architecture

Project Assignment

Project Title:

Design and Implementation of an Enterprise Application

Overview:

In this project, students will work on designing and partially implementing an enterprise application of their choice. The project is divided into milestones, each focusing on a key phase of enterprise system design and development. The goal is to help students understand the core architectural concepts, system modeling, and development practices used in real-world enterprise applications.

Milestone 1: Select an Enterprise Application (Due: 01/05/2026)

Each team must select an enterprise application to work on throughout the course. The application should be substantial enough to include multiple components, entities, and services.

Example Enterprise Application:

Food Delivery System

This system will manage customer accounts, restaurant listings, and menu items, while also handling complex business processes such as order placement, payment processing, delivery assignment, order tracking, and transaction history, going far beyond basic CRUD operations.

Students are required to design their selected enterprise application using a **Micro Services Architecture**, where the system is decomposed into independent, loosely coupled services.

Other Sample Ideas:

- E-commerce Platform (separate services for orders, payments, inventory, delivery)
- Online Banking System
- Supply Chain Management System
- Learning Management System (LMS)
- Video Streaming Platform (user service, recommendation service, streaming service)
- Airline Reservation System

Students must submit a short proposal (1-2 paragraphs) explaining:

- Main Functional Areas of the Organization
- Main Services/Components (decomposed into independent, loosely coupled services)
- Explain why these services are required
- Intended users

Milestone 2: System Modeling (Due: 22/05/2026)

Objective:

Draw the following diagrams to model the core structure and behavior of your selected enterprise application.

Tasks:

1. **Component Diagram:**

Create Component diagram showing how these components interact with each other to complete system functions. This diagram helps understand the overall system structure. Example: The component diagram shows the main parts of the Food Delivery Platform such as Customer Service, Restaurant Service, Order Service, Payment Service, and Delivery Service.

2. **Sequence Diagram:**

Create sequence diagrams showing the interaction between users and system components for a selected set of functionalities. The sequence diagram shows the step-by-step interaction between users and system components when an action occurs, such as placing an order. It explains how requests move from the customer to services like order, payment, and delivery. This helps visualize the workflow of the system.

3. **Use Case Diagram:**

The use case diagram shows the different users of the system like Customer, Restaurant Owner, Delivery Agent, and Admin. It describes the main functions each user can perform such as ordering food, managing menus, and tracking deliveries. This diagram focuses on system requirements from a user perspective.

4. **Deployment Diagram:**

The deployment diagram shows how the system is physically deployed across servers and devices. It includes user applications, API gateway, microservices, databases, and external systems like payment and maps services. This diagram explains how the system runs in a real environment.

Submission Requirements:

- UML diagrams must be created using a standard diagramming tool (e.g., Lucidchart, Draw.io, StarUML, etc.)
- A short description (3-5 lines) explaining each diagram

Milestone 3: REST API Design and Implementation (Due: 12/06/2026)

Objective:

Assuming that each microservice is implemented as a RESTful web service, you are required to design and develop 3 to 5 core services of the application using modern frameworks and technologies.

Note: It is acceptable to use the same technology stack for all services.

Tasks:

1. Identify **3 or 5 key micro-services** from your application.
2. For each service, design and implement RESTful API endpoints.

Note: There is no requirement to develop an API Gateway for this application; you may implement and test each microservice independently.

Examples of service wise:

1. Customer Service

Endpoint	HTTP Method	Description	Technology Used
/api/customers	GET	Retrieve all customers	Spring Boot, MySQL
/api/customers/{id}	GET	Retrieve customer by ID	Spring Boot, MySQL
/api/customers	POST	Register a new customer	Spring Boot, MySQL
/api/customers/{id}	PUT	Update customer details	Spring Boot, MySQL

2. Order Service

Endpoint	HTTP Method	Description	Technology Used
/api/orders	POST	Place a new order	Spring Boot, PostgreSQL
/api/orders/{id}	GET	Track order status	Spring Boot, PostgreSQL
/api/orders/{id}	PUT	Update order status	Spring Boot, PostgreSQL
/api/orders/{id}	DELETE	Cancel an order	Spring Boot, PostgreSQL

3. Restaurant Service

Endpoint	HTTP Method	Description	Technology Used
/api/restaurants	GET	Retrieve all restaurants	Node.js (Express.js), MongoDB
/api/restaurants/{id}	GET	Retrieve restaurant details	Node.js (Express.js), MongoDB
/api/menu	POST	Add new menu item	Node.js (Express.js), MongoDB
/api/menu/{id}	DELETE	Remove menu item	Node.js (Express.js), MongoDB

Submission Requirements:

- A Word/PDF document listing all endpoints with explanations.
- Include screenshots of testing the API endpoints using tools such as Postman or Swagger.
- Source code (push to GitHub or upload as ZIP).

Grading Breakdown:

Component	Weight
Application Proposal & Selection	10%
UML Diagrams (Usecase+ Component+Deployment+ Sequence)	30%
REST API Design and Implementation	40%
Technical Report and Documentation	20%

Notes:

- Students may work in teams of 4.
- Plagiarism will result in zero marks and academic penalties.
- Submit all milestones on the course LMS.